Programming in Python Midterm Project Report

Project Name: Inventory & Billing System

Course Name: Programming in Python

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Section: C

Semester: Spring 25-26

1. Overview:

The project is titled Inventory & Billing System. In this, desktop application inventory for a super shop can be managed as well as billing for an order and generating subsequent invoice can be done.

2. Track:

Title	Details
Track	Track 2 — Desktop App (GUI)
External Libraries	Customtkinter, Pillow
Language	Python 3 (Python 3.14.3)
Storage	JSON [for storing and saving data (store_data.json)], CSV([for exporting invoice data (invoice_history.csv)]

3. Setup & Run Instructions:

3.1 Environment Setup:

Step1: Create venv environment:

```
py -m venv .venv
```

or

```
python3 -m venv .venv
```

Step2: Activate venv environment:

Windows: `.venv\Scripts\activate`

MacOS/Linux: `source .venv/bin/activate`

Step3: Install dependencies:

```
python -m pip install -r requirements.txt
```

Step 4: Run the application:

```
python main.py
```

3.2 Dependencies (Listed in requirements.txt):

customtkinter==5.2.2

darkdetect==0.8.0

packaging==26.0

pillow==12.1.1

3.3 Additional Install:

Install Custom Fonts:

Go to Assets Folder and install:

"Caveat-Regular.ttf"

“Manrope-ExtraBold.ttf"

"Poppins-Regular.ttf"

Now, the application should run as Intended.

4. Feature List (Screenshots are included):

I. Dashboard with two different views (Inventory Management & Billing & Invoice System)

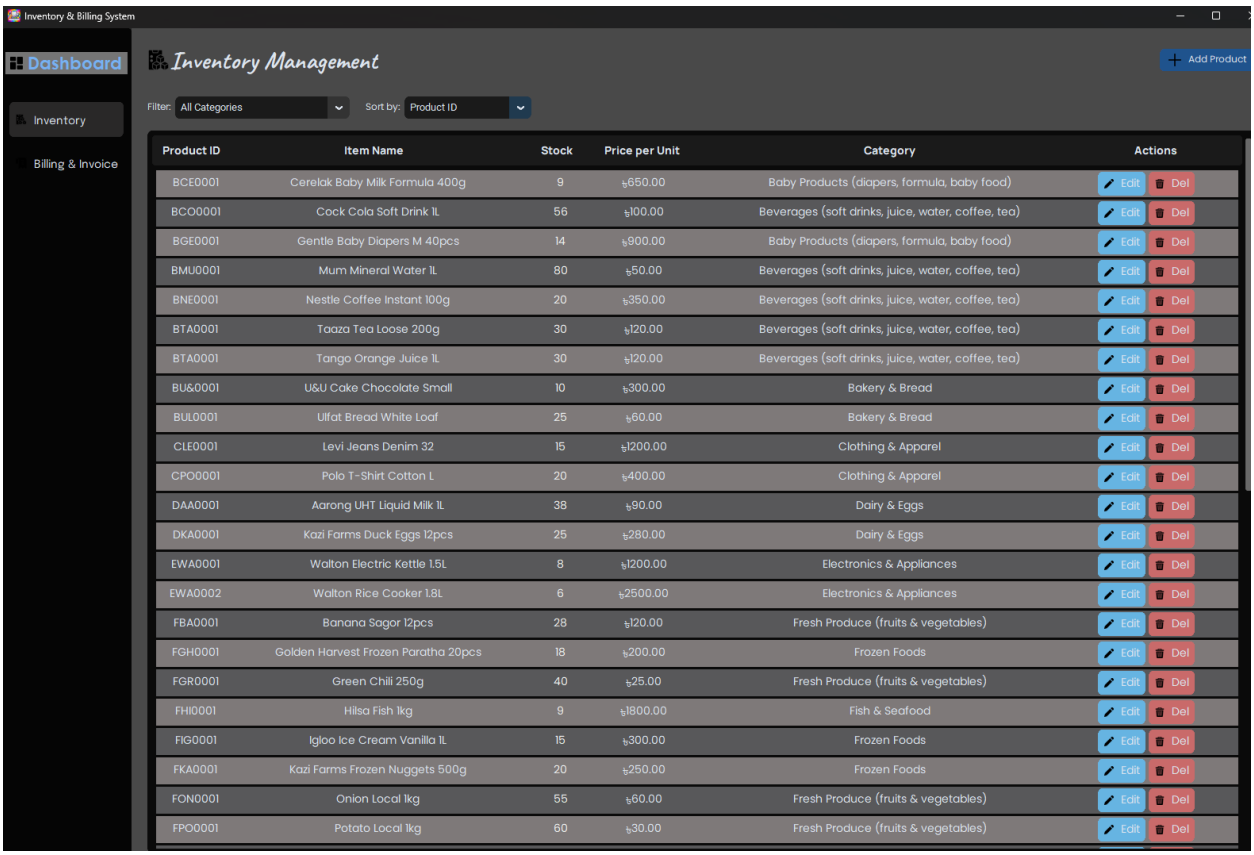


Fig. 1: Inventory Management View

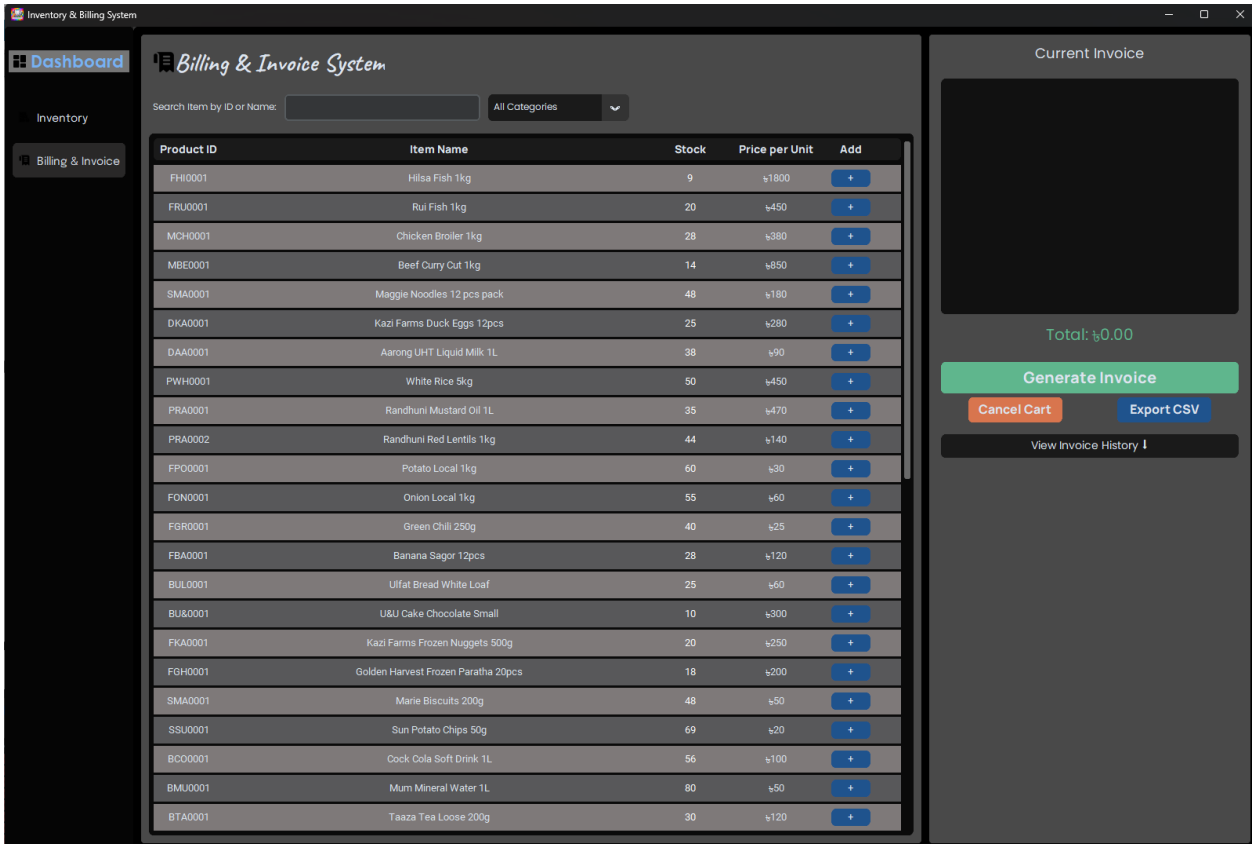


Fig. 2: Billing & Invoice System View

II. Add New Product to Inventory:

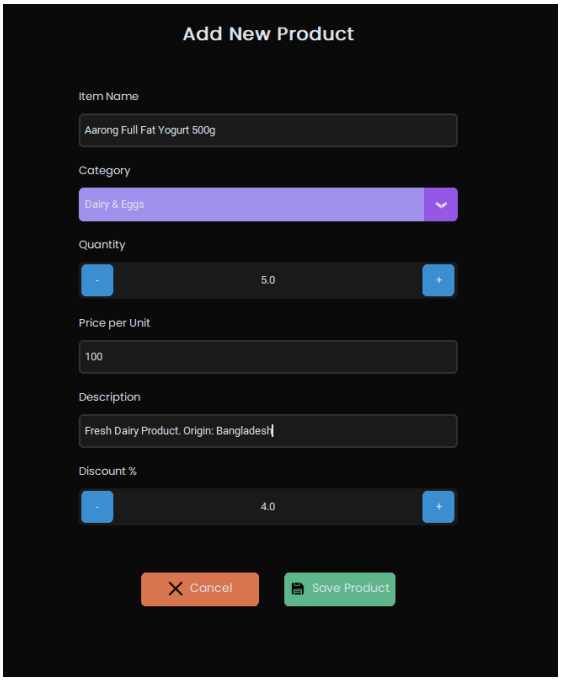


Fig. 3: Adding New Product, giving details

Product ID	Item Name	Stock	Price per Unit	Category	Actions
DAA0002	Aarong Full Fat Yogurt 500g	5 (!)	৳100.00	Dairy & Eggs	Edit Del

Fig. 4: New Product Added

III. Edit Existing Product in Inventory:

DAA0001	Aarong UHT Liquid Milk 1L	38	৳90.00	Dairy & Eggs	Edit Del

Fig. 5: Product Details before Editing

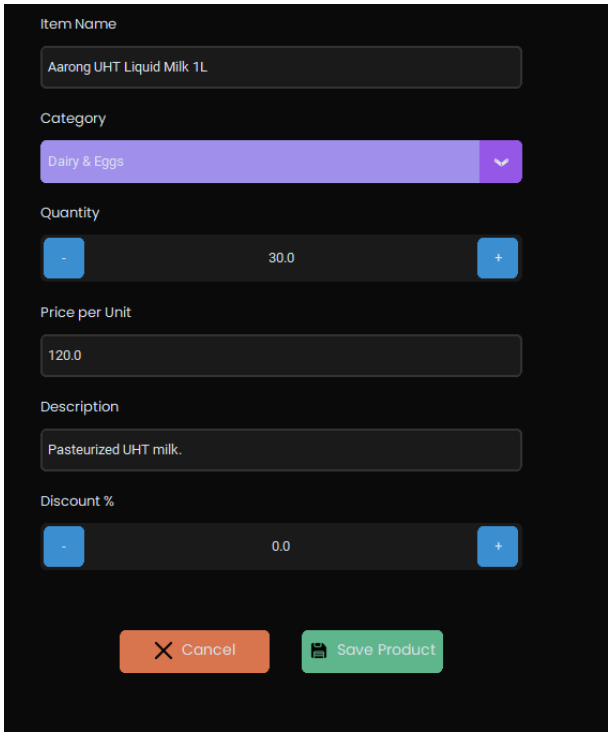


Fig. 6: Edited some Product Details

DAA0001	Aarong UHT Liquid Milk 1L	30	৳120.00	Dairy & Eggs	Edit Del

Fig. 7: Product Details after Editing

IV. Deleting an already existing product from inventory:

Product ID	Item Name	Stock	Price per Unit	Category	Actions
DAA0002	Aarong Full Fat Yogurt 500g	5 (!)	৳100.00	Dairy & Eggs	Edit Del

Fig. 8: Product exists before deleting

Product ID	Item Name	Stock	Price per Unit	Category	Actions
DAA0001	Aarong UHT Liquid Milk 1L	30	৳120.00	Dairy & Eggs	Edit Del
FBA0001	Banana Sagor 12pcs	28	৳120.00	Fresh Produce (fruits & vegetables)	Edit Del
OBA0001	Bashundhara Notebook A4 100pg	50	৳80.00	Office & School Supplies	Edit Del
HBA0001	Basundhara Toilet Tissue 4roll	30	৳120.00	Household Essentials (cleaning supplies, paper goods)	Edit Del

Fig. 9: Product doesn’t exist after Editing

V. Custom Sorting via any column:

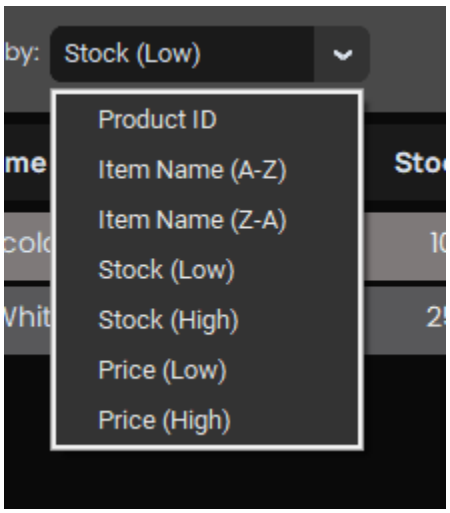
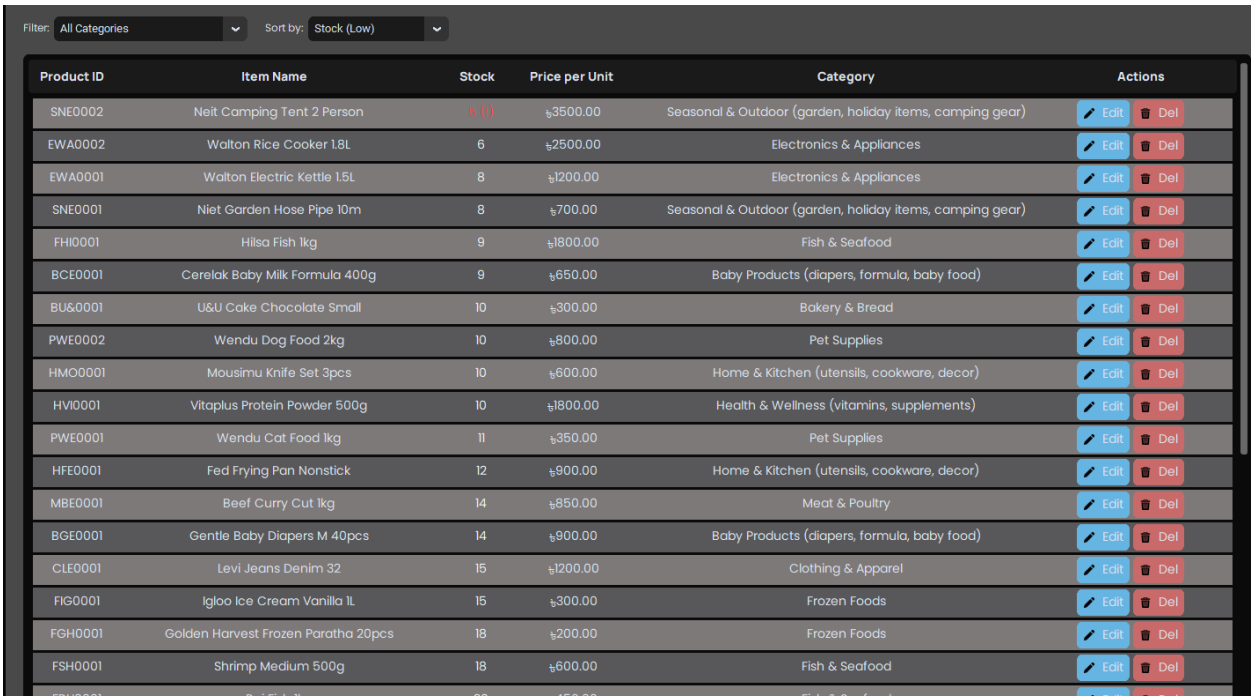


Fig. 9: All the available sortings



Filter: All Categories Sort by: Stock (Low)

Product ID	Item Name	Stock	Price per Unit	Category	Actions
SNE0002	Neit Camping Tent 2 Person	5 (!)	৳3500.00	Seasonal & Outdoor (garden, holiday items, camping gear)	Edit Del
EWA0002	Walton Rice Cooker 1.8L	6	৳2500.00	Electronics & Appliances	Edit Del
EWA0001	Walton Electric Kettle 1.5L	8	৳1200.00	Electronics & Appliances	Edit Del
SNE0001	Niet Garden Hose Pipe 10m	8	৳700.00	Seasonal & Outdoor (garden, holiday items, camping gear)	Edit Del
FHI0001	Hilsa Fish 1kg	9	৳1800.00	Fish & Seafood	Edit Del
BCE0001	Cerelak Baby Milk Formula 400g	9	৳650.00	Baby Products (diapers, formula, baby food)	Edit Del
BU&0001	U&U Cake Chocolate Small	10	৳300.00	Bakery & Bread	Edit Del
PWE0002	Wendu Dog Food 2kg	10	৳800.00	Pet Supplies	Edit Del
HMO0001	Mousimu Knife Set 3pcs	10	৳600.00	Home & Kitchen (utensils, cookware, decor)	Edit Del
HVI0001	Vitaplus Protein Powder 500g	10	৳1800.00	Health & Wellness (vitamins, supplements)	Edit Del
PWE0001	Wendu Cat Food 1kg	11	৳350.00	Pet Supplies	Edit Del
HFE0001	Fed Frying Pan Nonstick	12	৳900.00	Home & Kitchen (utensils, cookware, decor)	Edit Del
MBE0001	Beef Curry Cut 1kg	14	৳850.00	Meat & Poultry	Edit Del
BGE0001	Gentle Baby Diapers M 40pcs	14	৳900.00	Baby Products (diapers, formula, baby food)	Edit Del
CLE0001	Levi Jeans Denim 32	15	৳1200.00	Clothing & Apparel	Edit Del
FIG0001	Igloo Ice Cream Vanilla 1L	15	৳300.00	Frozen Foods	Edit Del
FGH0001	Golden Harvest Frozen Paratha 20pcs	18	৳200.00	Frozen Foods	Edit Del
FSH0001	Shrimp Medium 500g	18	৳600.00	Fish & Seafood	Edit Del

Fig. 10: Sorted via Stock (Low to High)

VI. Custom Filter with all available categories:

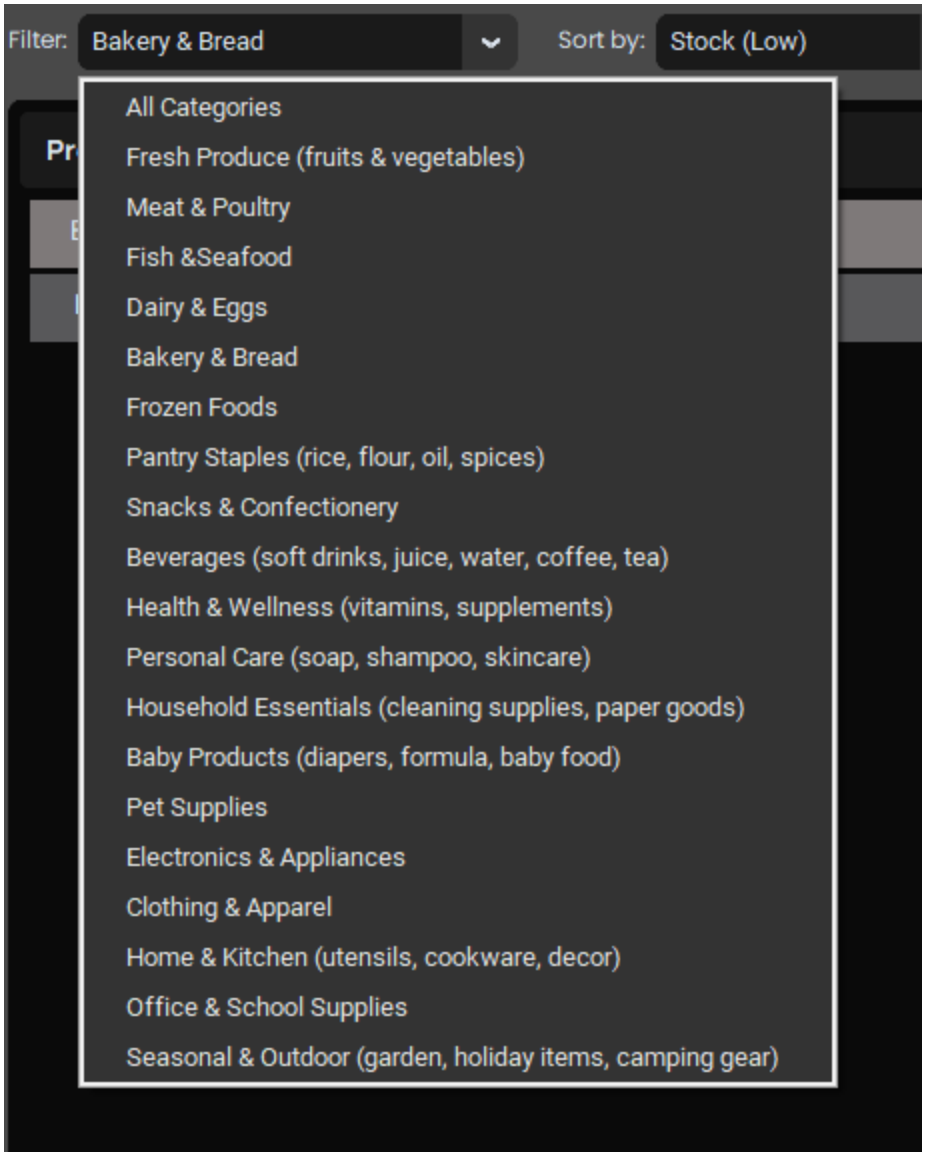


Fig. 11: All the available filters

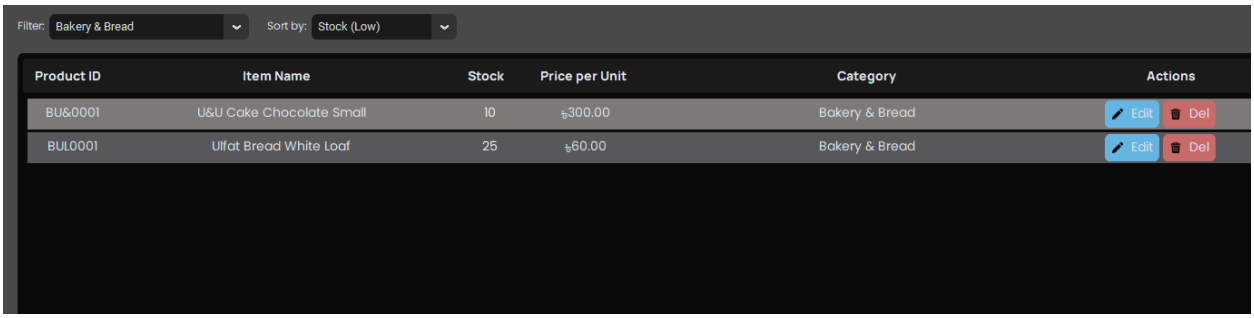


Fig. 12: Filtered by Bakery & Bread

VII. Custom Validation for Adding or Editing Product:

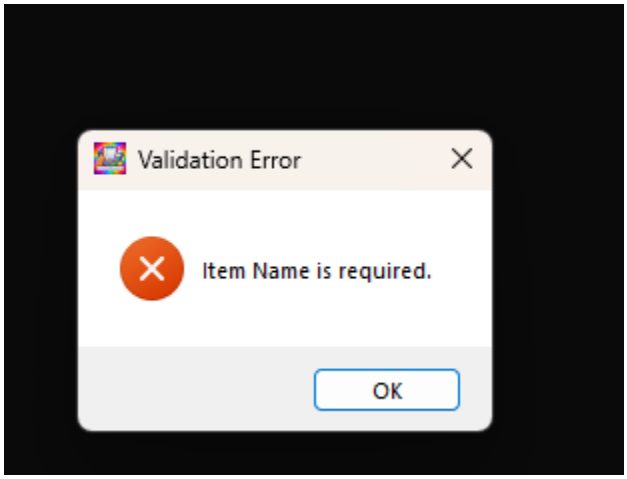


Fig. 13: No Product Name given

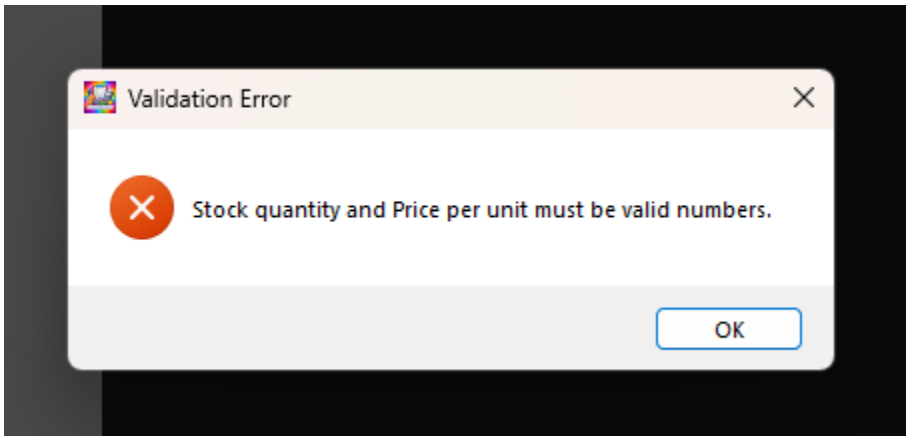


Fig. 14: Price per unit or stock quantity not properly given

IX. Searching Product by Product ID or Product Name:

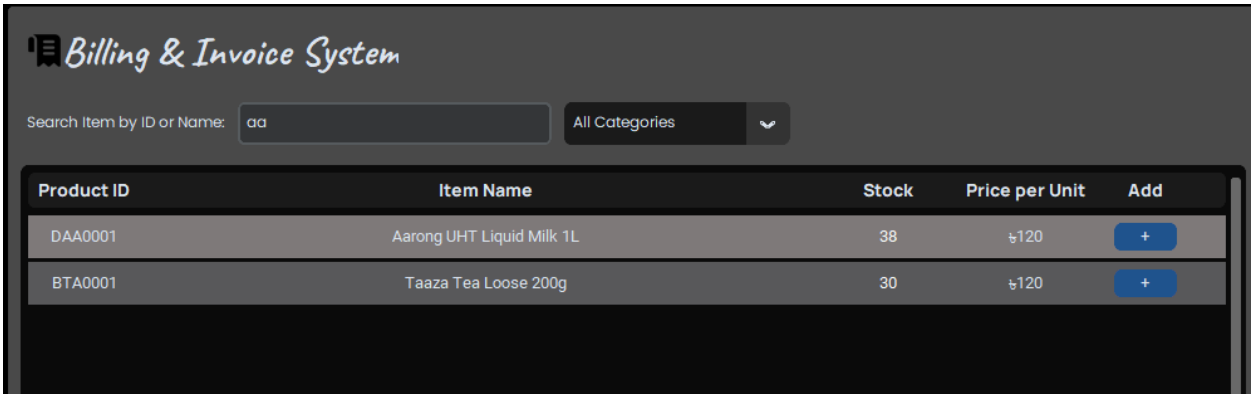


Fig. 14: Searched products containing aa in it's name

X. View current invoice after adding product to cart with all the calculations:

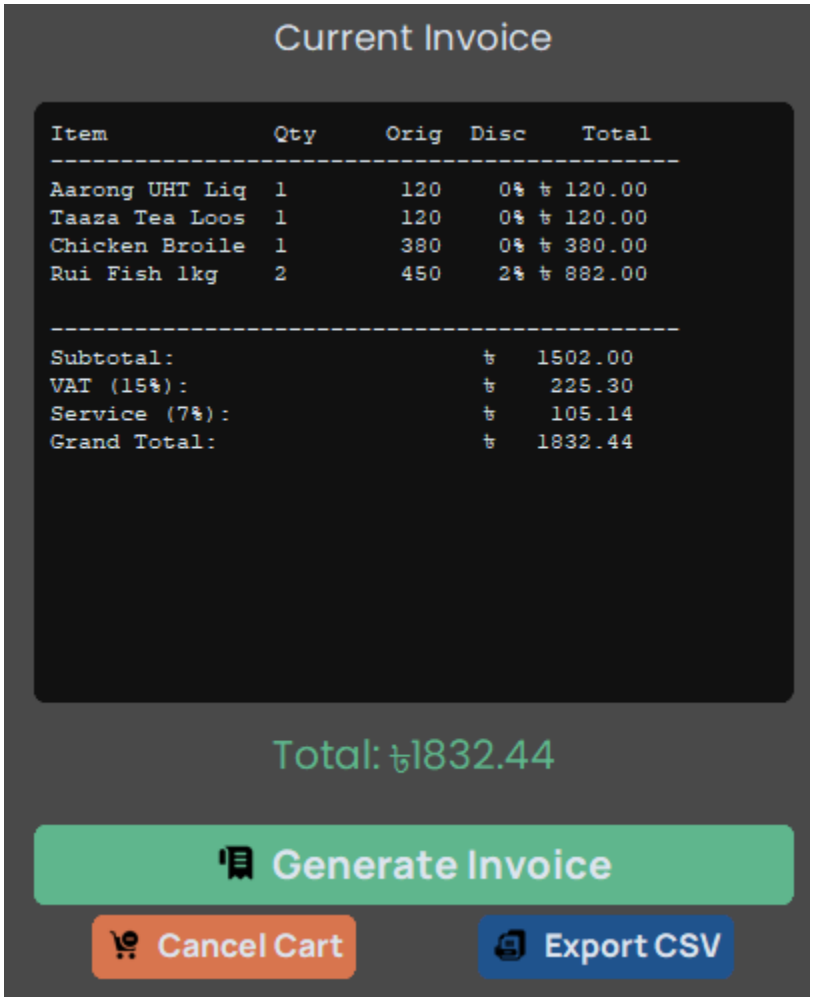


Fig. 15: Current Invoice View before generation.

XI. Generate Invoice and view previous invoices:

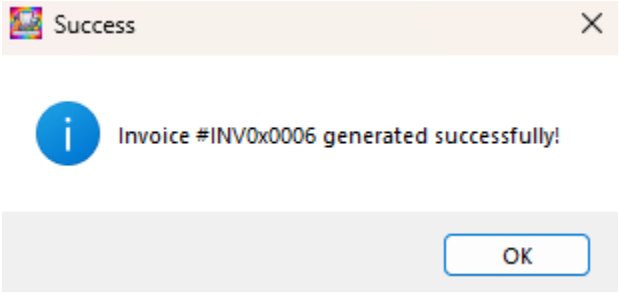


Fig. 16: Successful invoice generation message

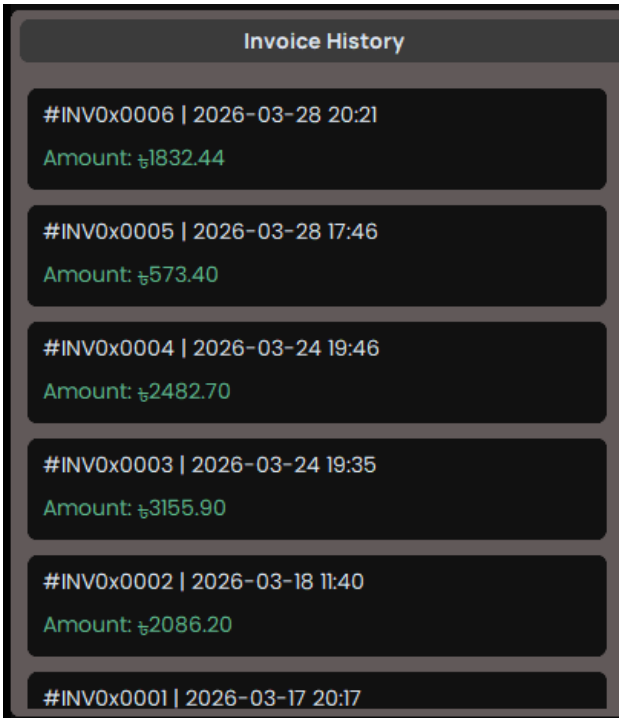


Fig. 17: View All invoices, latest first

XII. Cancelling current cart:

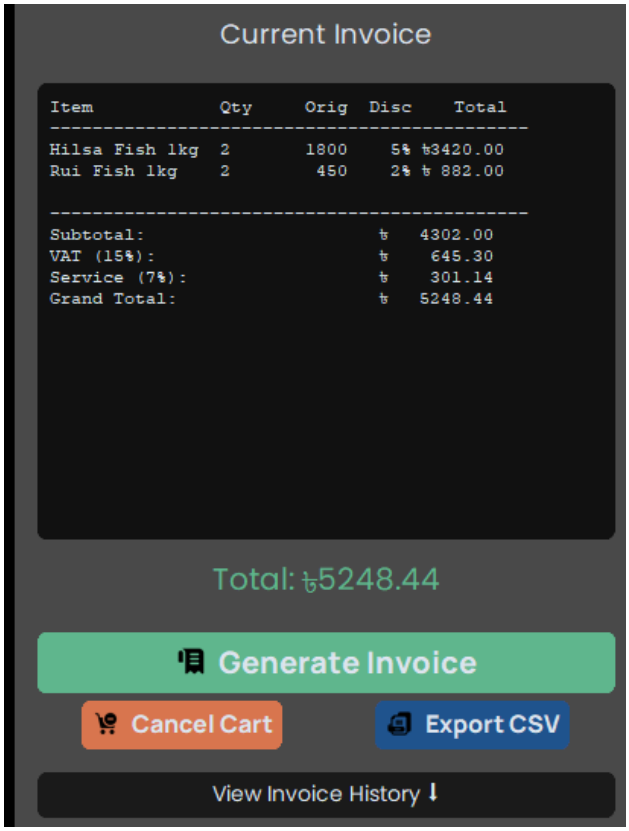


Fig. 18: View of a cart before confirming

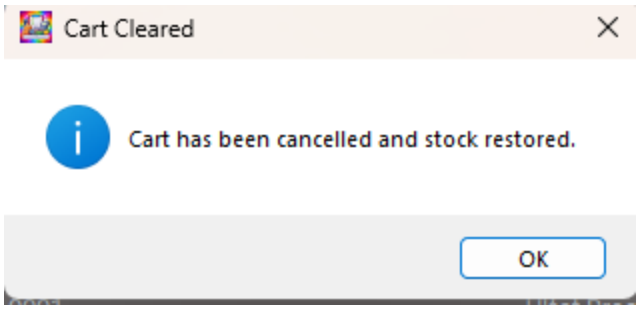


Fig. 19: Successful cart cancellation message

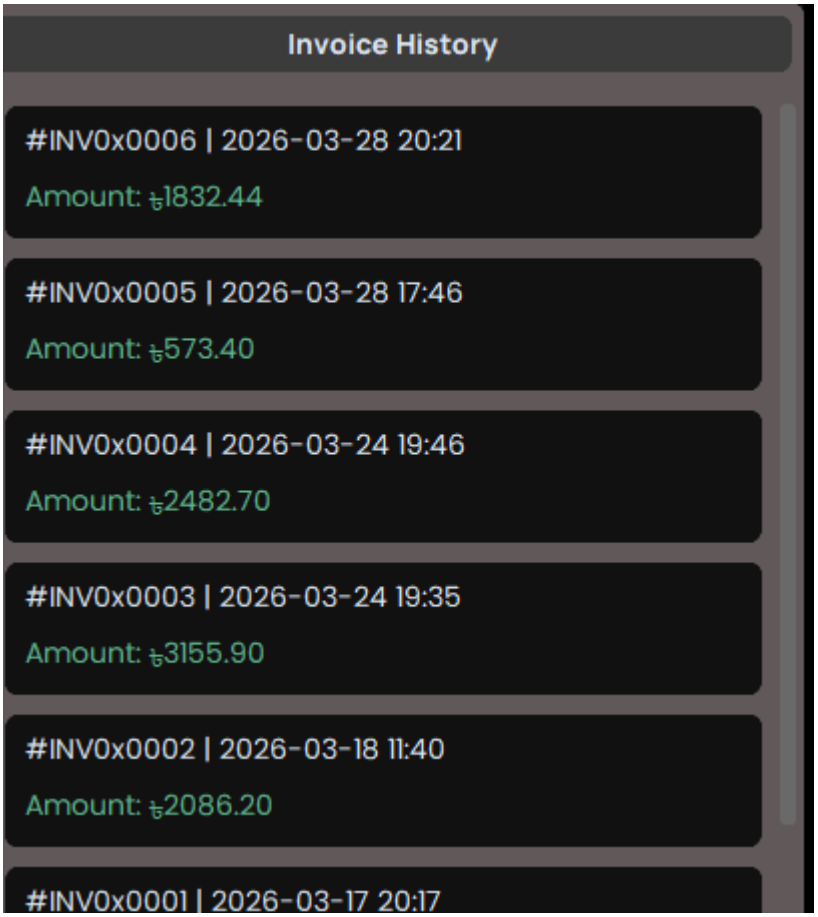


Fig. 20: No new invoice generated

XIII. Exporting Invoices to .csv file:

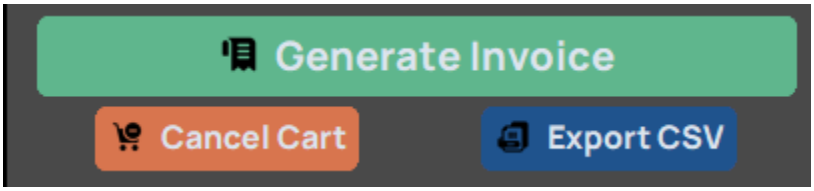


Fig. 21: Export all invoices to csv

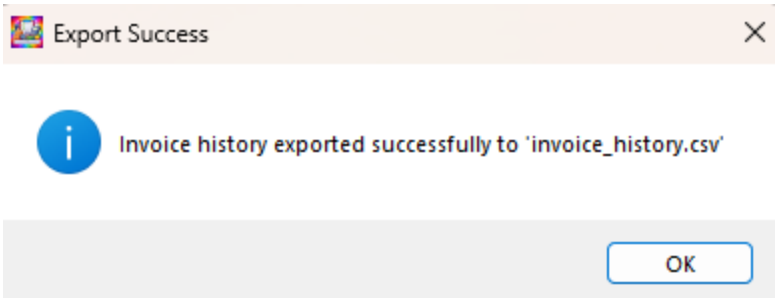
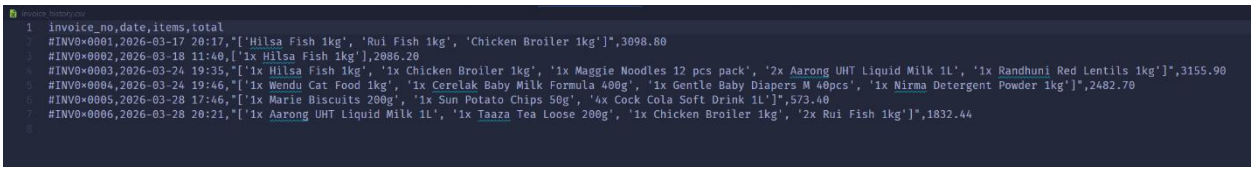


Fig. 22: All invoices exported successfully message



```
invoice_no,date,items,total
#INV0×0001,2026-03-17 20:17,"['Hilsa Fish 1kg', 'Rui Fish 1kg', 'Chicken Broiler 1kg']",3098.80
#INV0×0002,2026-03-18 11:40,['1x Hilsa Fish 1kg'],2086.20
#INV0×0003,2026-03-24 19:35,"['1x Hilsa Fish 1kg', '1x Chicken Broiler 1kg', '1x Maggie Noodles 12 pcs pack', '2x Aarong UHT Liquid Milk 1L', '1x Randhuni Red Lentils 1kg']",3155.90
#INV0×0004,2026-03-24 19:46,"['1x Wendu Cat Food 1kg', '1x Cerelak Baby Milk Formula 400g', '1x Gentle Baby Diapers M 40pcs', '1x Nirma Detergent Powder 1kg']",2482.70
#INV0×0005,2026-03-28 17:46,"['1x Marie Biscuits 200g', '1x Sun Potato Chips 50g', '4x Cock Cola Soft Drink 1L']",573.40
#INV0×0006,2026-03-28 20:21,"['1x Aarong UHT Liquid Milk 1L', '1x Taaza Tea Loose 200g', '1x Chicken Broiler 1kg', '2x Rui Fish 1kg']",1832.44
```

Fig. 23: Exported data in invoice_history.csv file